Random Forest Self-Training: Do Confidence Thresholds Affect Accuracy

Name: Anusha Baddam **Id: 24127511**

GitHub: Random_Forest_Self-Training-_Do_Confidence_Thresholds_Affect_Accuracy.git

Introduction:

Random Forest is a supervised machine learning algorithm that builds many decision trees and combines their outputs to make a final prediction. Each tree sees a random subset of the training data and, at each split, a random subset of the features. For classification, every tree votes for a class and the forest predicts the class with the most votes. For regression, it averages the predictions from all trees. Because the trees are different from each other, their individual mistakes tend to be cancled out. This makes the overall model more stable and less likely to overfit than a single deep decision tree. Random Forests can handle non-linear decision boundaries and usually work well with only a few main hyperparameters, such as the number of trees and the maximum depth.

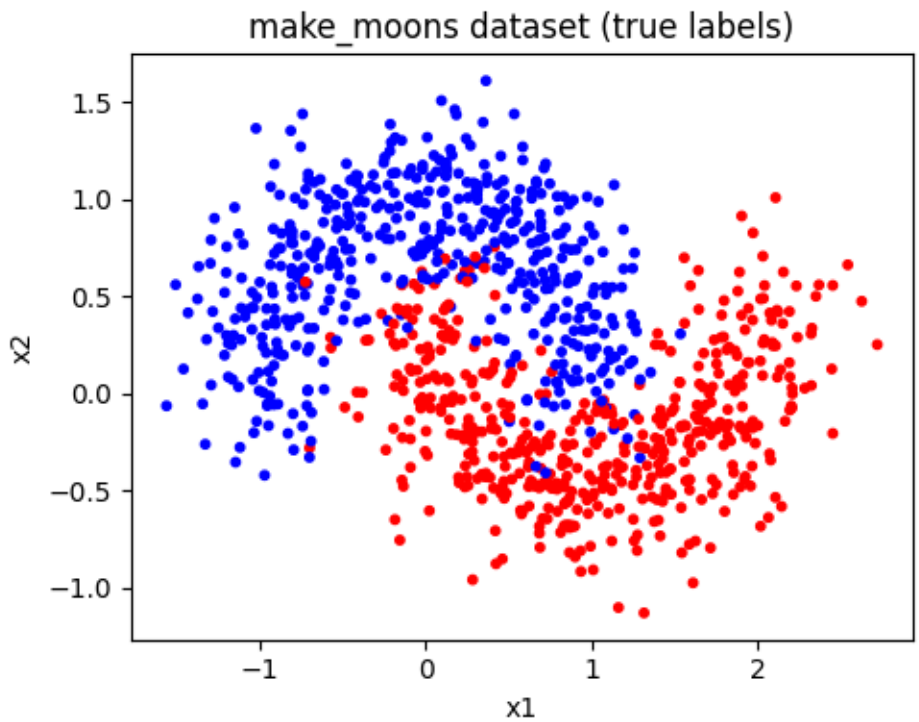
In many real problems it is expensive to label data, so we often have a small, labeled set and a much larger unlabeled set. Semi-supervised learning tries to use both labeled and unlabeled data together. Self-training is one of the simplest semi-supervised methods. In self-training with Random Forest, we first train the model only on the small, labeled subset. We then use this model to predict class probabilities for unlabeled examples. Points where the model is very confident (above a chosen probability threshold) are given pseudo-labels and added to the labeled set. The Random Forest is retrained on this expanded dataset, and the process can be repeated for several rounds.

The confidence threshold plays an important role in this process. A low threshold adds many pseudo-labels, but some of them may be wrong, which can hurt test accuracy. A high threshold adds fewer but more reliable pseudo-labels, so the model is safer but may learn less from the unlabeled data. This tutorial explores how changing that threshold affects the performance of a self-training Random Forest.

Methodology:

In this tutorial, we show how to use self-training with a Random Forest on a random two-class problem. We start by generating the make moons dataset from scikit-learn with 1,000 samples and some noise, which creates two interlocking half-moons that are not linearly separable.

Each point has two features $x1$ and $x2$ and belongs to one of two classes shown as blue and red clusters in the scatter plot, so the decision boundary must curve to follow the moon shapes rather than being a straight line. We then split the data into 80% training and 20% test sets, keeping the class balance the same.



To mimic a semi-supervised setting, we randomly pick 100 training points and keep their labels, and we treat the remaining 700 training points as unlabeled. First, we train a normal Random Forest classifier with 200 trees only on the 100 labeled points and measure its test accuracy. This acts as our supervised baseline.

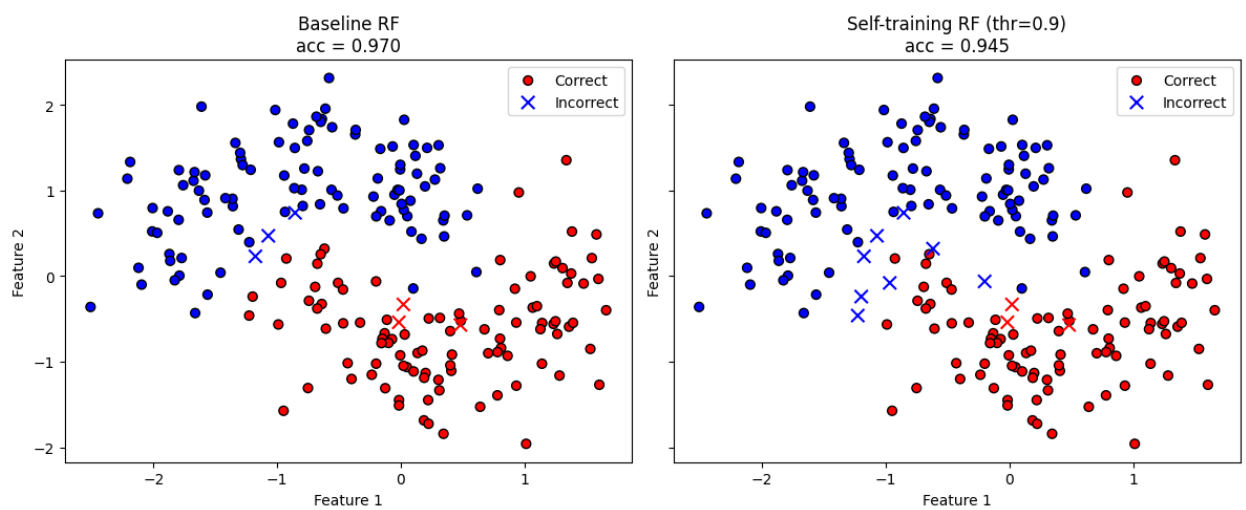
Next, we turn this into a self-training loop. In each round, we fit a Random Forest on the current labeled set, use it to predict class probabilities for all unlabeled points, and then select only those points where the model is very confident (for example, maximum class probability above 0.8, 0.9, or 0.95). We give these points pseudo-labels from the model, add them to the labeled set, and remove them from the unlabeled pool. We repeat this process until no more points pass the threshold or we reach 10 iterations and we then compare the final test accuracy for each threshold with the original baseline.

Results:

On this dataset, the Random Forest trained only on the 100 labeled points already achieves high test accuracy (around 0.97), so it is a strong supervised baseline. When we apply self-training with a relatively low threshold such as 0.8, the model quickly adds many pseudo-labeled points from the unlabeled pool, but some of these labels are incorrect, so the test accuracy can fluctuate or

slightly decrease compared to the baseline. With a threshold of 0.9, fewer points are added in each iteration, but the pseudo-labels are more reliable and the final accuracy remains close to the baseline while using more data. For a very high threshold like 0.95, the model becomes conservative, adding only a small number of highly confident points; accuracy is stable but there is little gain from the unlabeled data because the labeled set grows slowly. These experiments illustrate that the confidence threshold controls a trade-off between using more unlabeled data and avoiding noisy pseudo-labels.

When we compare the test predictions of the baseline Random Forest and the self-training Random Forest with a threshold of 0.9. On both plots, red circles and blue circles show correctly classified points from the two classes, while blue crosses mark the few misclassified points. The baseline model achieves accuracy 0.970, slightly higher than the self-training model's 0.945, which suggests that on this simple dataset self-training with this threshold does not improve performance over the already strong supervised model.



How the Threshold Effects Accuracy and the Model?

The confidence threshold mainly affects which pseudo-labels are added, so it changes both accuracy and model behaviour's.

- With a low threshold (for example 0.8), many unlabeled points are added at each iteration, including points the model is not very sure about. This quickly increases the size of the labeled set, but it also introduces more wrong pseudo-labels, which can slightly reduce or destabilise test accuracy as these errors get reinforced. The model's decision boundary may become more irregular because it tries to fit noisy labels.
- With a medium threshold (around 0.9), fewer points are added per iteration, but they are more likely to be correct. Accuracy stays close to the supervised baseline while still benefiting from extra data, and the decision boundary is refined in a more stable way.

- With a very high threshold (for example 0.95), the model only adds very confident points, so almost all pseudo-labels are correct but the labeled set grows slowly. Accuracy is safe (rarely worse than the baseline), yet there is limited improvement because most unlabeled data are never used.

Takeaways:

These experiments show that there is no single "best" threshold; instead, it depends on whether we care more about using as much unlabeled data as possible or about keeping labels very clean. On our simple make moons example, the supervised Random Forest baseline is already strong, so self-training with aggressive (low) thresholds can slightly harm accuracy by introducing noisy pseudo-labels, while moderate or high thresholds keep performance close to the baseline but offer only small gains. In harder real-world problems where the initial labeled set is weaker, carefully tuning the threshold is likely to matter more, because it controls how quickly the model can "teach itself" without drifting due to wrong labels.

Conclusion:

In this tutorial, we used a Random Forest with self-training to explore how the confidence threshold controls the use of unlabeled data on a simple make moons task. The baseline Random Forest trained on only 100 labeled points already reached high accuracy, and self-training with a threshold of 0.9 produced a slightly lower accuracy while adding extra pseudo-labeled points. This shows that when the supervised model is already strong and the dataset is relatively easy, aggressive use of unlabeled data does not necessarily help and can even hurt if noisy pseudo-labels are added. In more challenging problems with fewer labels or more complex structures, carefully tuning the threshold is likely to be more beneficial, as it balances learning from additional data against the risk of propagating errors.

References:

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